

*Skip – List*  
*–Search Element*  
*–Program Analysis*

**Program Analysis:**

```
void searchElement(struct SkipList *lst, int key) {
    struct Node *current = lst->header;

    // Traverse from highest level to the lowest
    for (int i = lst->level; i >= 0; i--) {
        while (current->forward[i] != NULL && current->forward[i]->key <
key) {
            current = current->forward[i];
        }
    }

    // Step into the exact match at level 0
    current = current->forward[0];

    if (current != NULL && current->key == key) {
        cout << "Found key " << key << " in the Skip List." << endl;
    } else {
        cout << "Key " << key << " not found." << endl;
    }
}

...
case 3:
    cout << "Enter key to search: ";
    cin >> key;
    searchElement(list, key);
    break;
```

## Program Analysis

There are three cases:

1<sup>st</sup>, when  $Current = NULL$  [ List  
is  
Empty ]

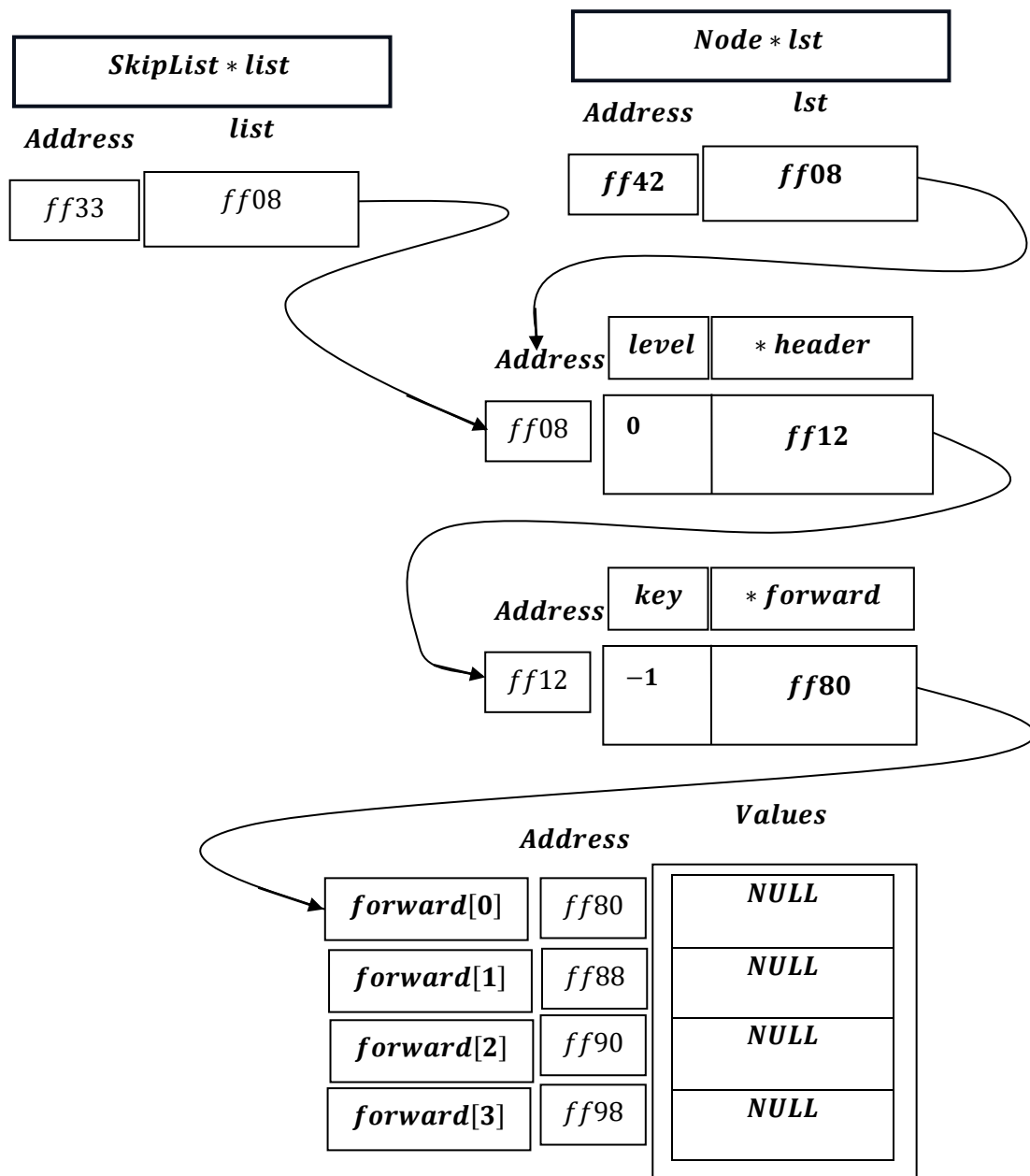
```
searchElement(list, key);
```

$searchElement(list: ff08, key: 20) \rightarrow passed.$

And,

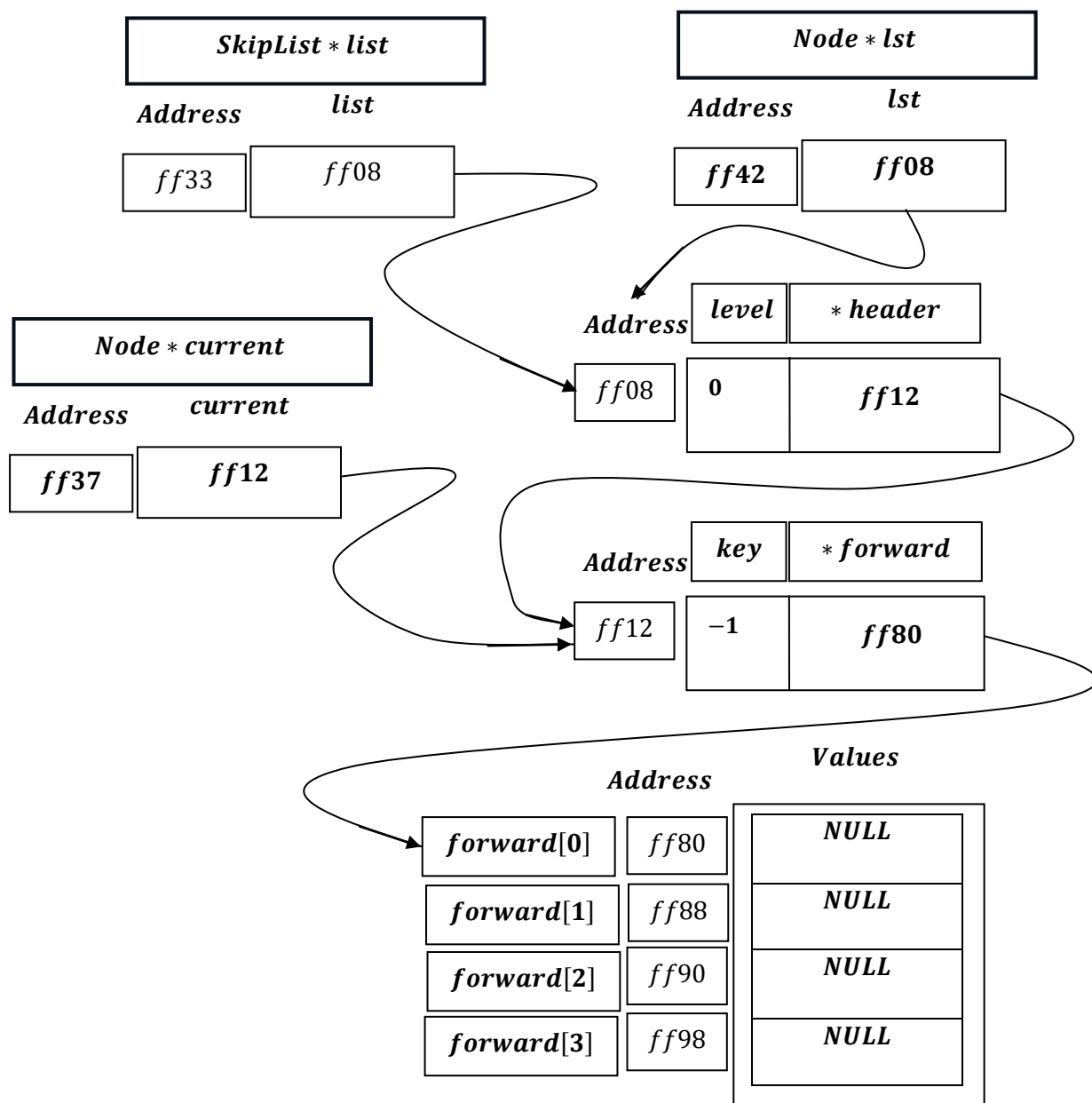
```
void searchElement(struct SkipList *lst, int key)
```

$searchElement(list := lst: ff08, key := key: 20) \rightarrow Assigned$



```
struct Node *current = lst->header;
```

*current = lst → header = ff12.*



```
for (int i = lst->level; i >= 0; i--)
```

$i = \text{lst} \rightarrow \text{level} = 0$  and  $i \geq 0$  [Condition is True]

```
while (current->forward[i] != NULL && current->forward[i]->key < key)
```

$\text{current} \rightarrow \text{forward}[0]$  i.e.  $\text{ff80} = \text{NULL}$ ., hence condition is false .

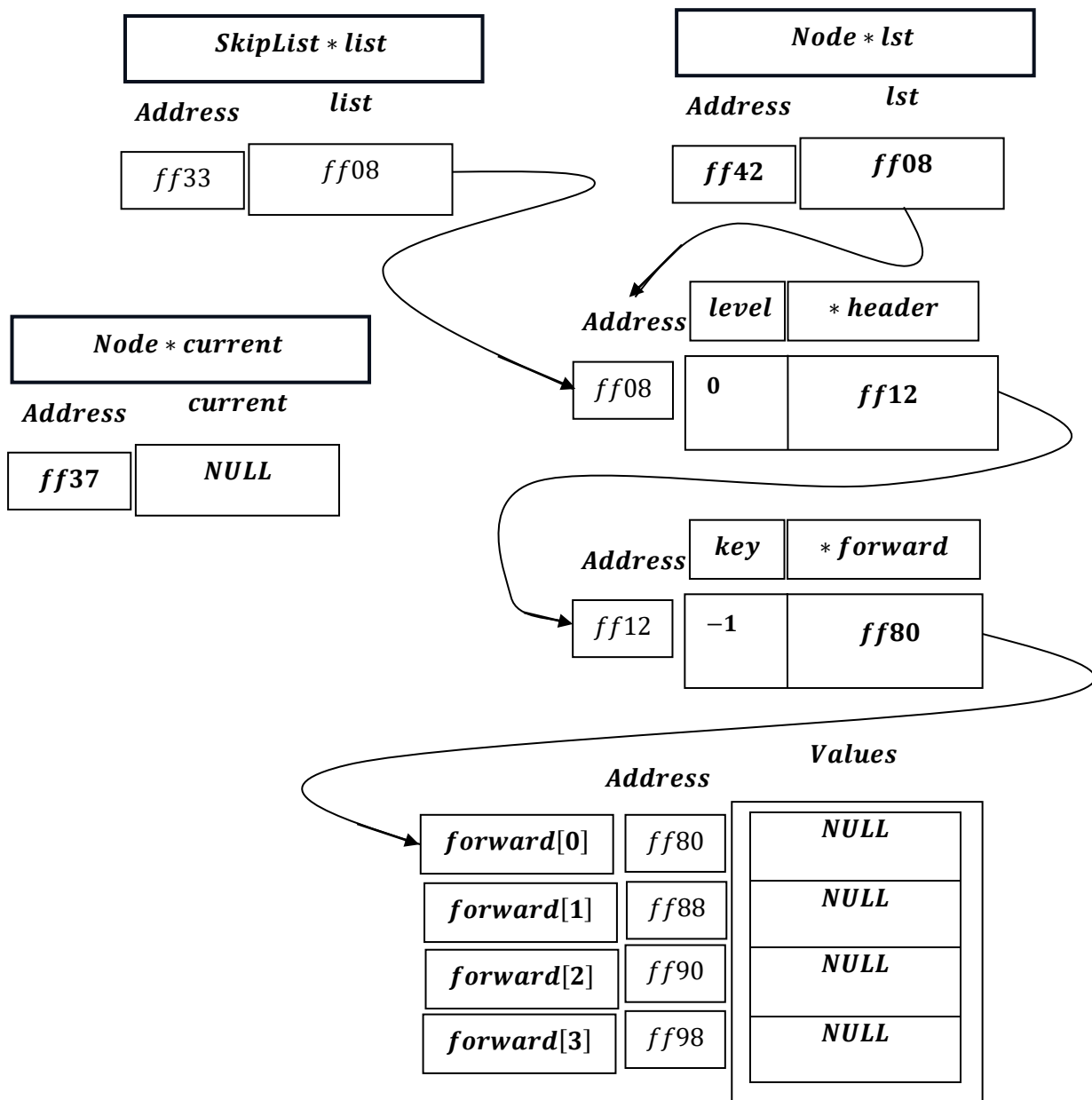
$i-- \Rightarrow i = i - 1 \Rightarrow i = 0 - 1 = -1$ .

```
for (int i = lst->level; i >= 0; i--)
```

$i = -1$  and  $i \geq 0$   $\left[ \begin{array}{c} \text{Condition} \\ \text{Fails} \\ \text{and} \\ \text{Loop Exits} \end{array} \right]$

```
current = current->forward[0];
```

$\text{current} = \text{current} \rightarrow \text{forward}[0]$  i.e.  $\text{ff80} = \text{NULL}$



```
if (current != NULL && current->key == key)
```

*Current*  $\neq$  *NULL* [Condition Fails, hence if exits].

```
else {
    cout << "Key " << key << " not found." << endl;
}
```

*Hence, now it prints ``Key 20 not found.`` and function exits.*

---

*2<sup>nd</sup>, when Current  $\neq$  NULL  $\left[ \begin{array}{l} \text{List} \\ \text{is not} \\ \text{Empty} \end{array} \right]$*

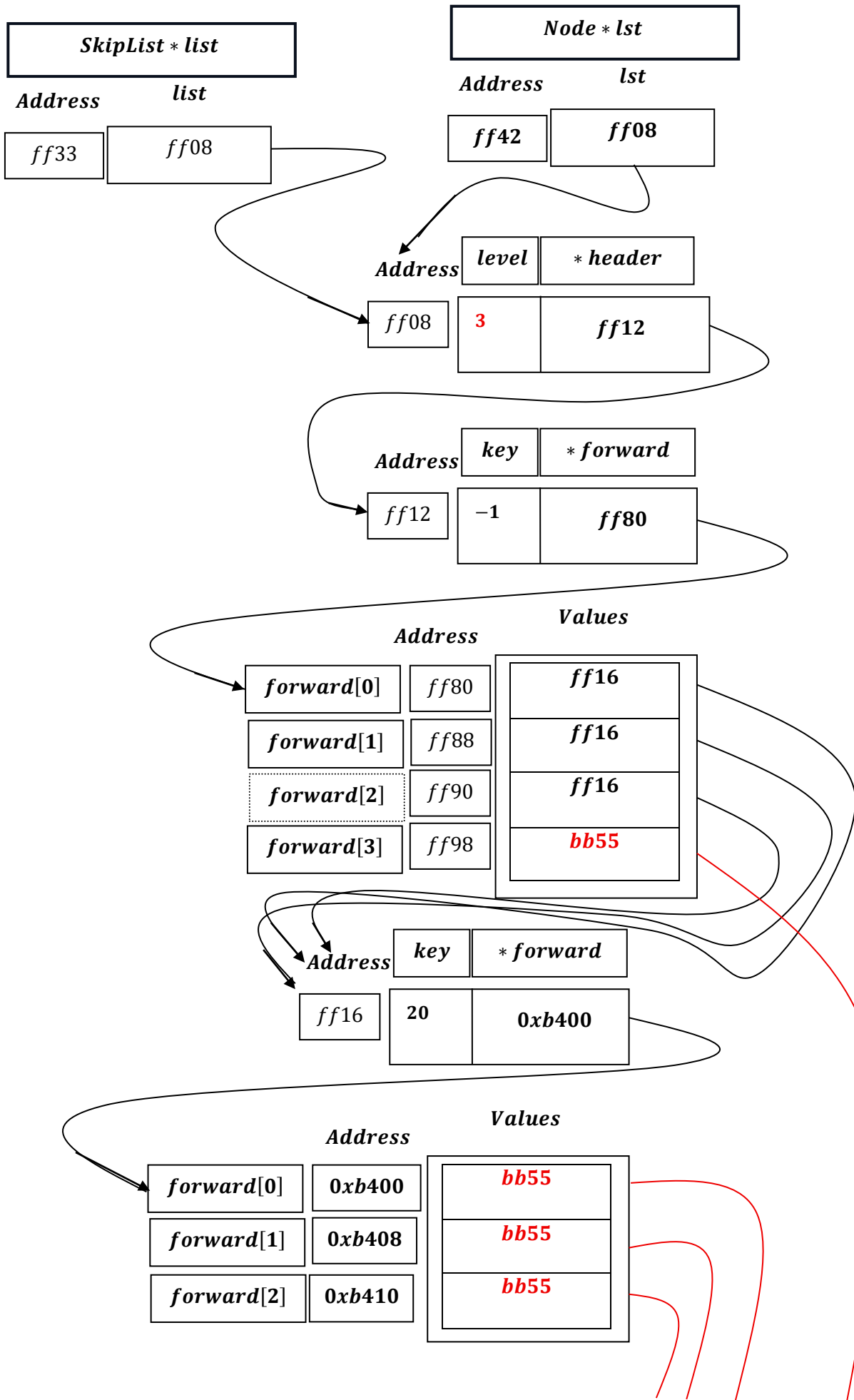
```
searchElement(list, key);
```

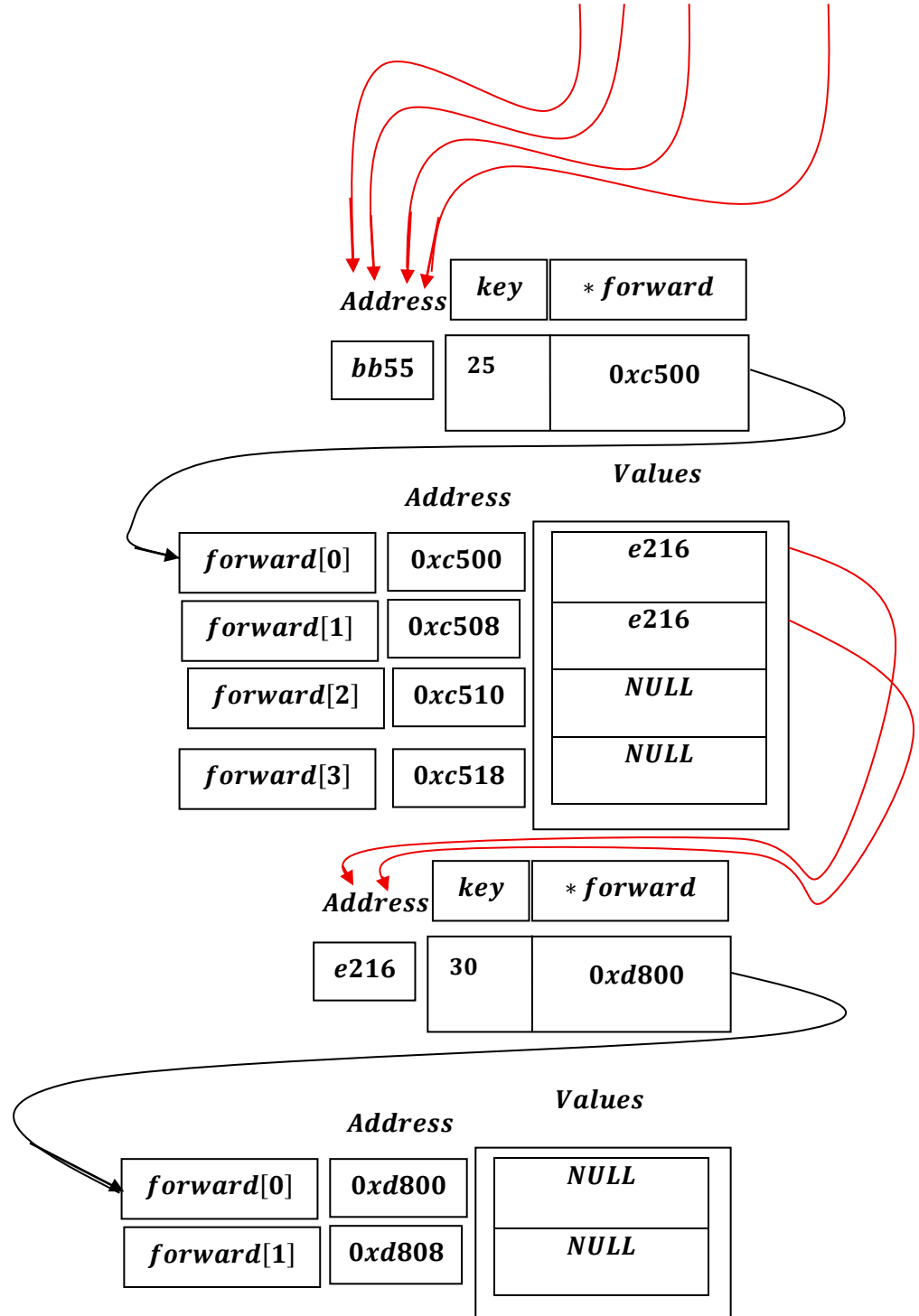
*searchElement(list: ff08, key: 30)  $\rightarrow$  passed.*

*And,*

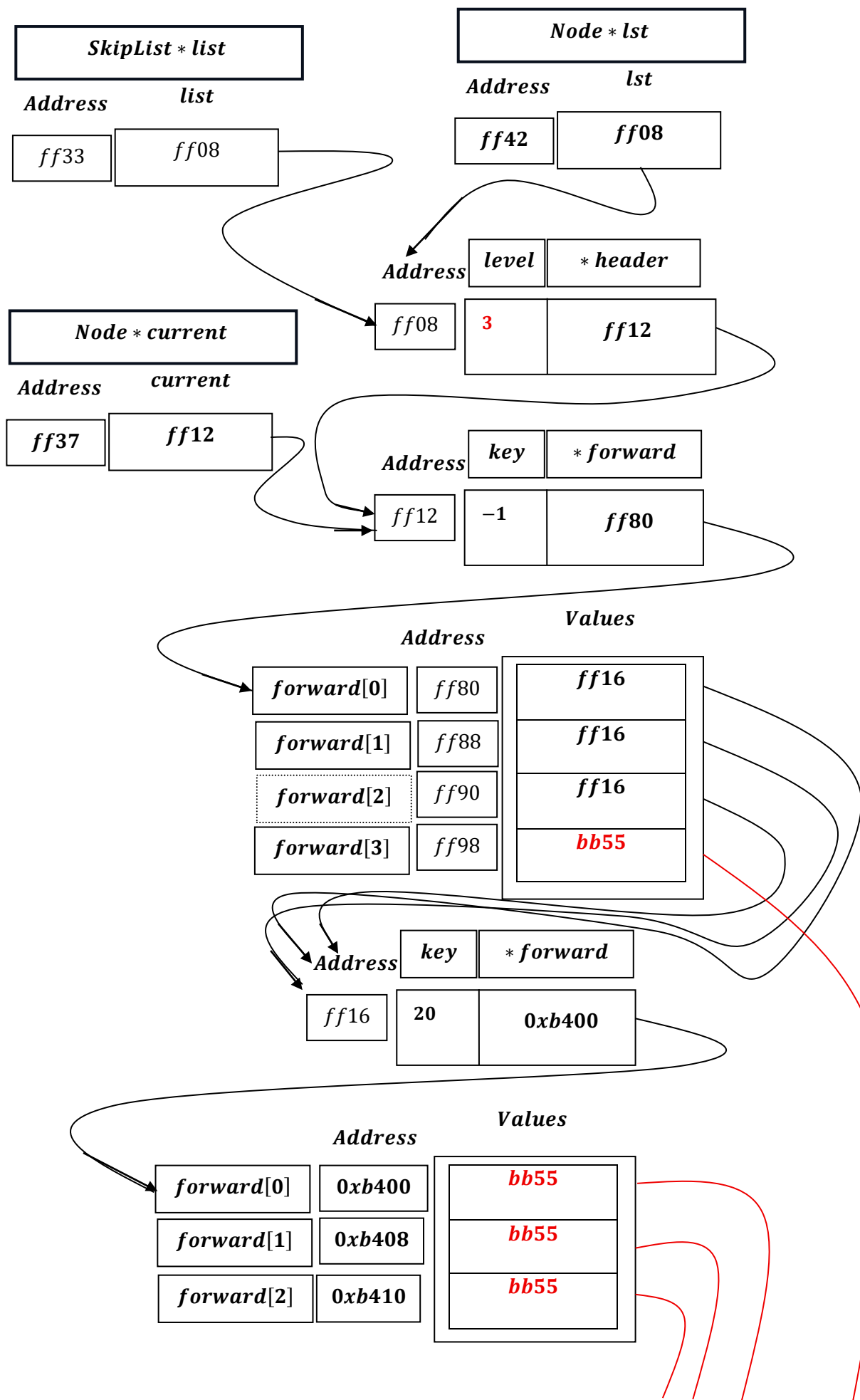
```
void searchElement(struct SkipList *lst, int key)
```

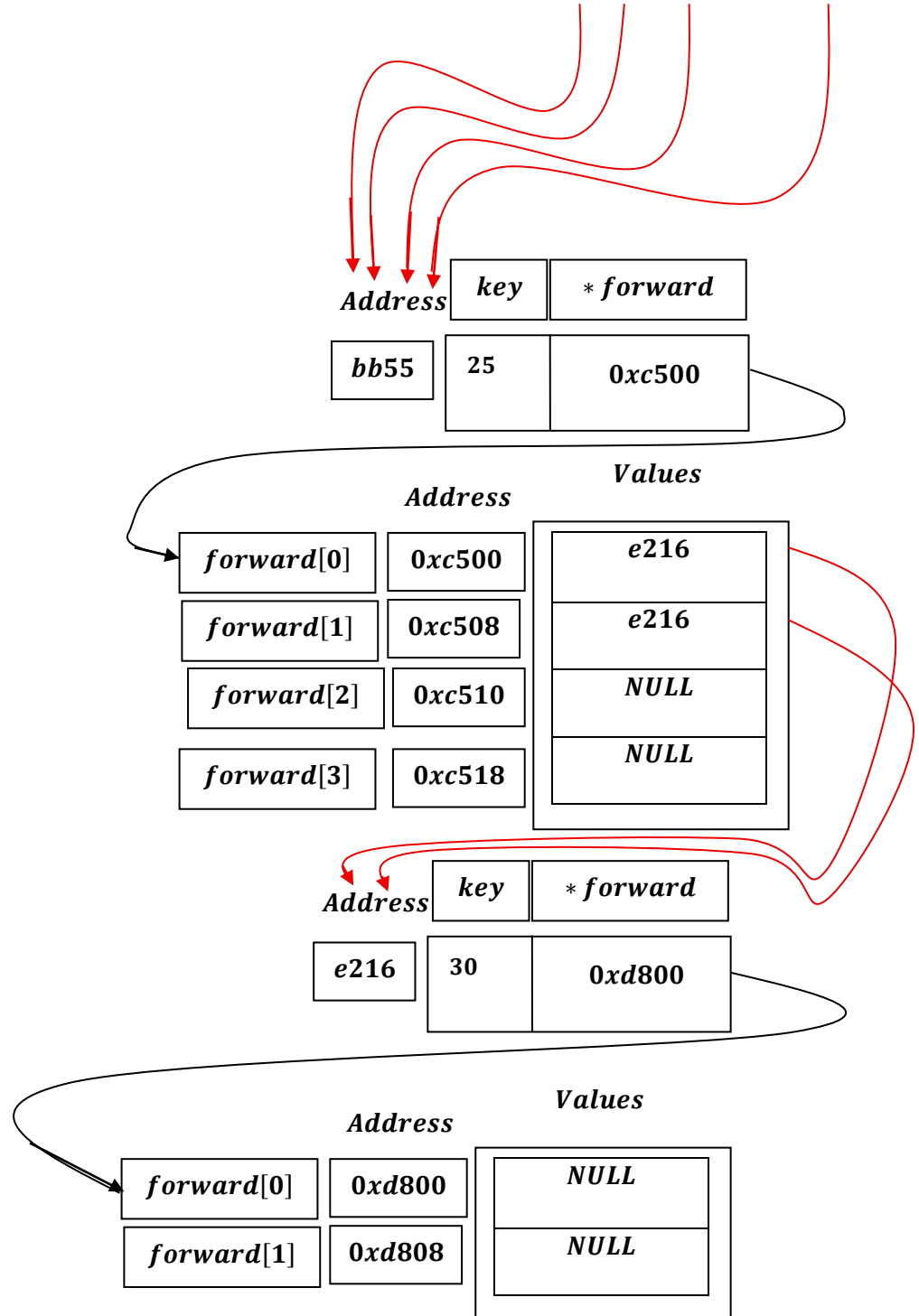
*searchElement(list := lst: ff08, key := key: 30)  $\rightarrow$  Assigned*





```
struct Node *current = lst->header;
current = lst → header = ff12.
```





```

for (int i = lst->level; i >= 0; i--) {
    while (current->forward[i] != NULL && current->forward[i]->key <
key) {
        current = current->forward[i];
    }
}

```

*lst* → *level* = 3

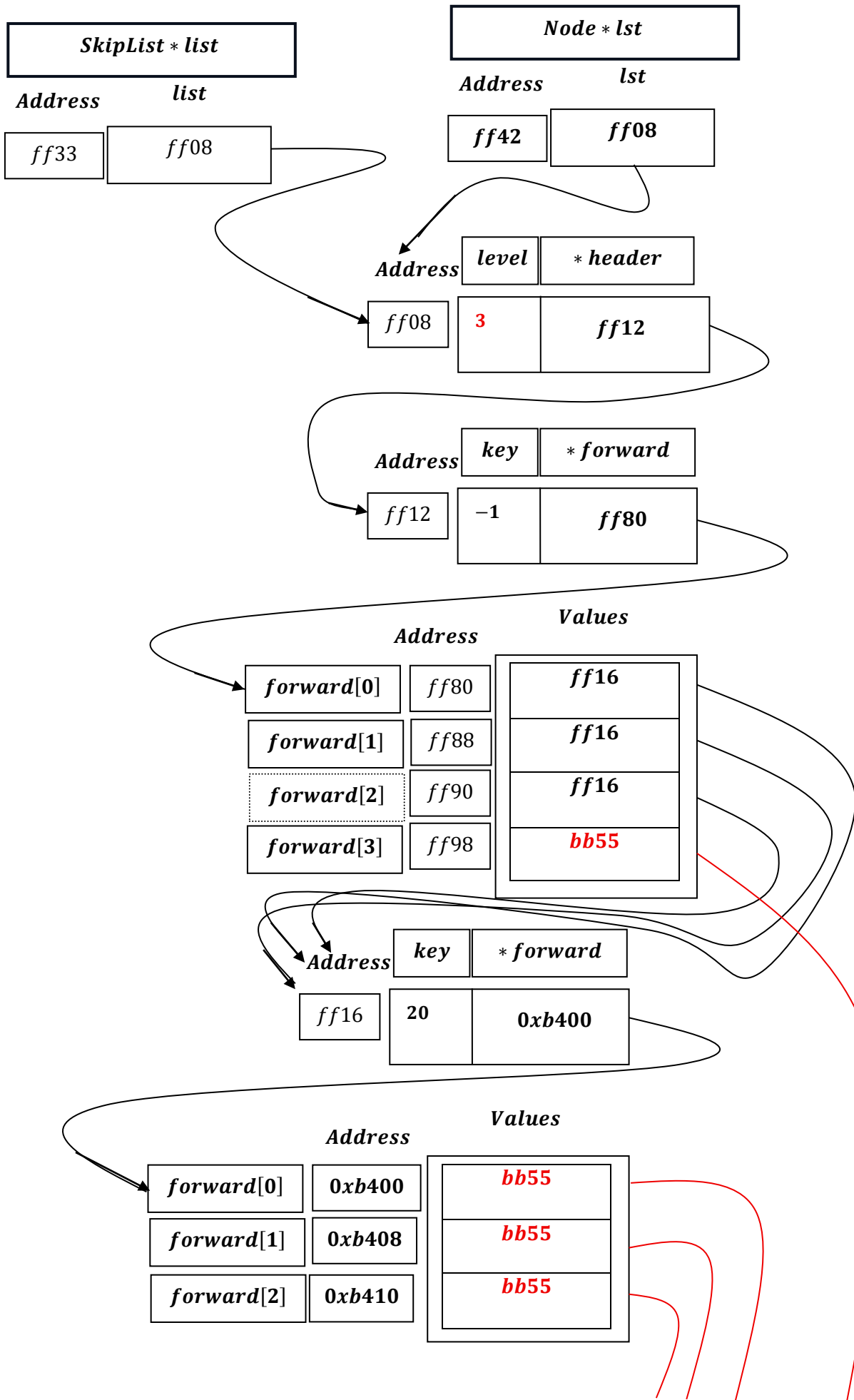
***$i = lst \rightarrow level = 3; i \geq 0$***

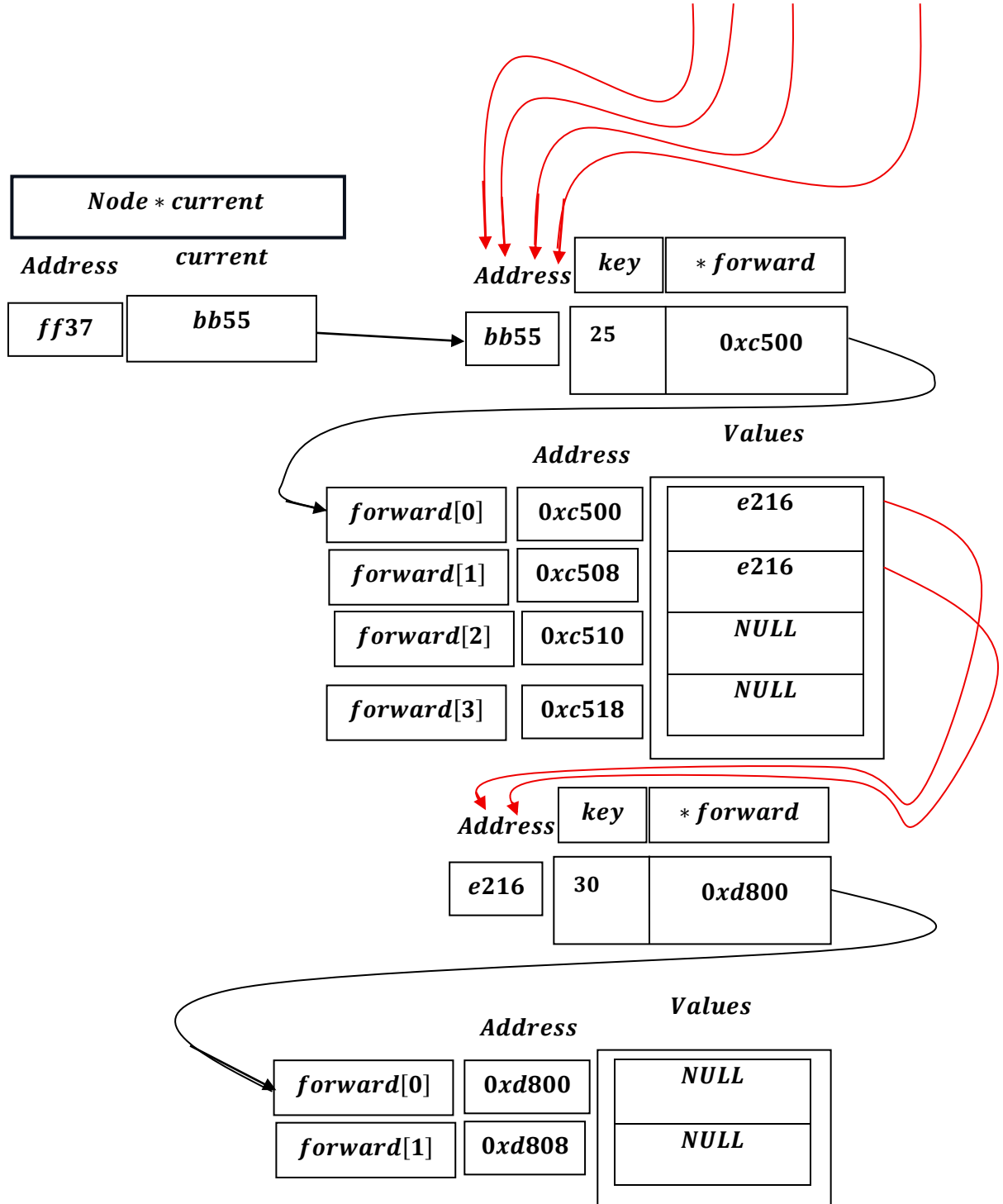
***while: current: ff12  $\rightarrow$  forward[3]: ff98  $\rightarrow$  bb55  $\neq$  NULL [True] **AND*****

***current: ff12  $\rightarrow$  forward[3]: ff98[bb55]  $\rightarrow$  key: 25  $<$  key: 30 [True]:***

***current = current: ff12  $\rightarrow$  forward[3]: ff98 = bb55***

***while: current: bb55  $\rightarrow$  forward[3]: 0xc518: NULL  $\neq$  NULL [False]***





$$i \longrightarrow i = i - 1 = 3 - 1 = 2$$

**$i = 2; i \geq 0$**

**$\text{while: current: } bb55 \rightarrow \text{forward}[2]: 0xc518: NULL \neq NULL \text{ [False]}$**

**$i -- \Rightarrow i = i - 1 = 2 - 1 = 1$**

**$i = 1; i \geq 0$**

**$\text{while: current: } bb55 \rightarrow \text{forward}[1]: 0xc508 \rightarrow e216 \neq NULL \text{ [True]}$**

**$\text{current: } bb55 \rightarrow \text{forward}[1]: 0xc508[e216] \rightarrow \text{key: } 30 < \text{key: } 30 \text{ [False]}$**

**$i -- \Rightarrow i = i - 1 = 1 - 1 = 0$**

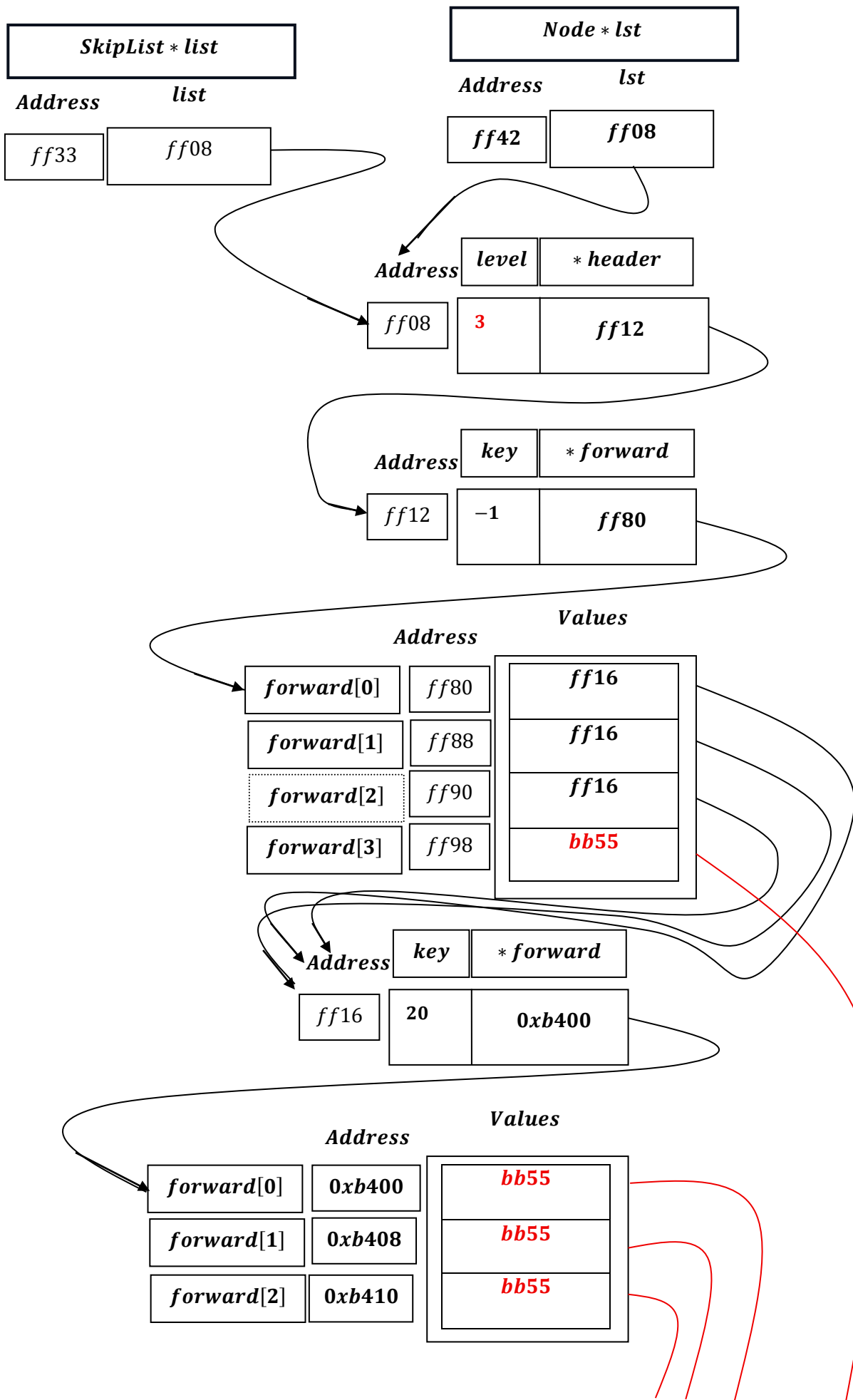
**$i = 0; i \geq 0$**

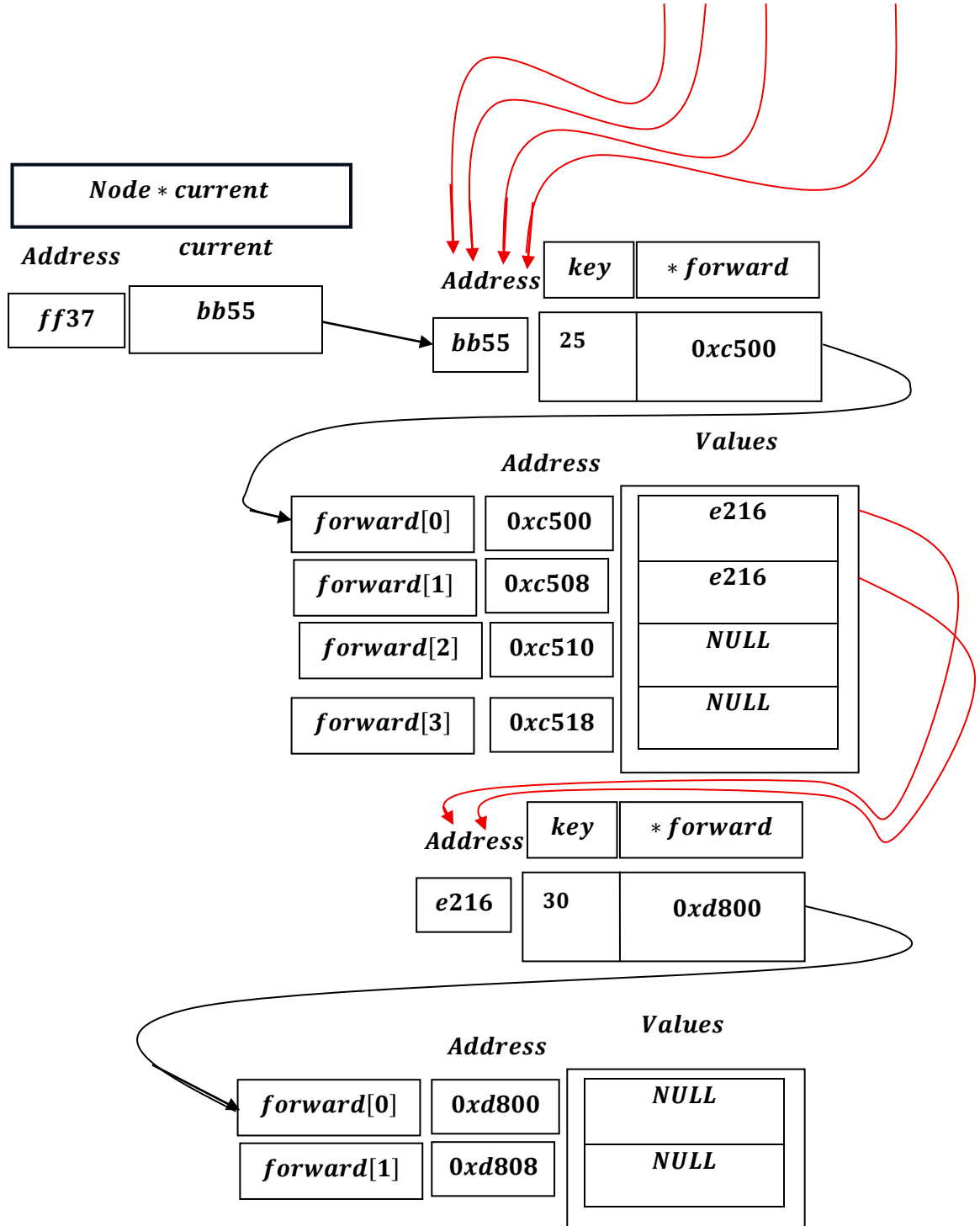
**$\text{while: current: } bb55 \rightarrow \text{forward}[0]: 0xc500 \rightarrow e216 \neq NULL \text{ [True]}$**

**$\text{current: } bb55 \rightarrow \text{forward}[0]: 0xc500[e216] \rightarrow \text{key: } 30 < \text{key: } 30 \text{ [False]}$**

**$i -- \Rightarrow i = i - 1 = 0 - 1 = -1$**

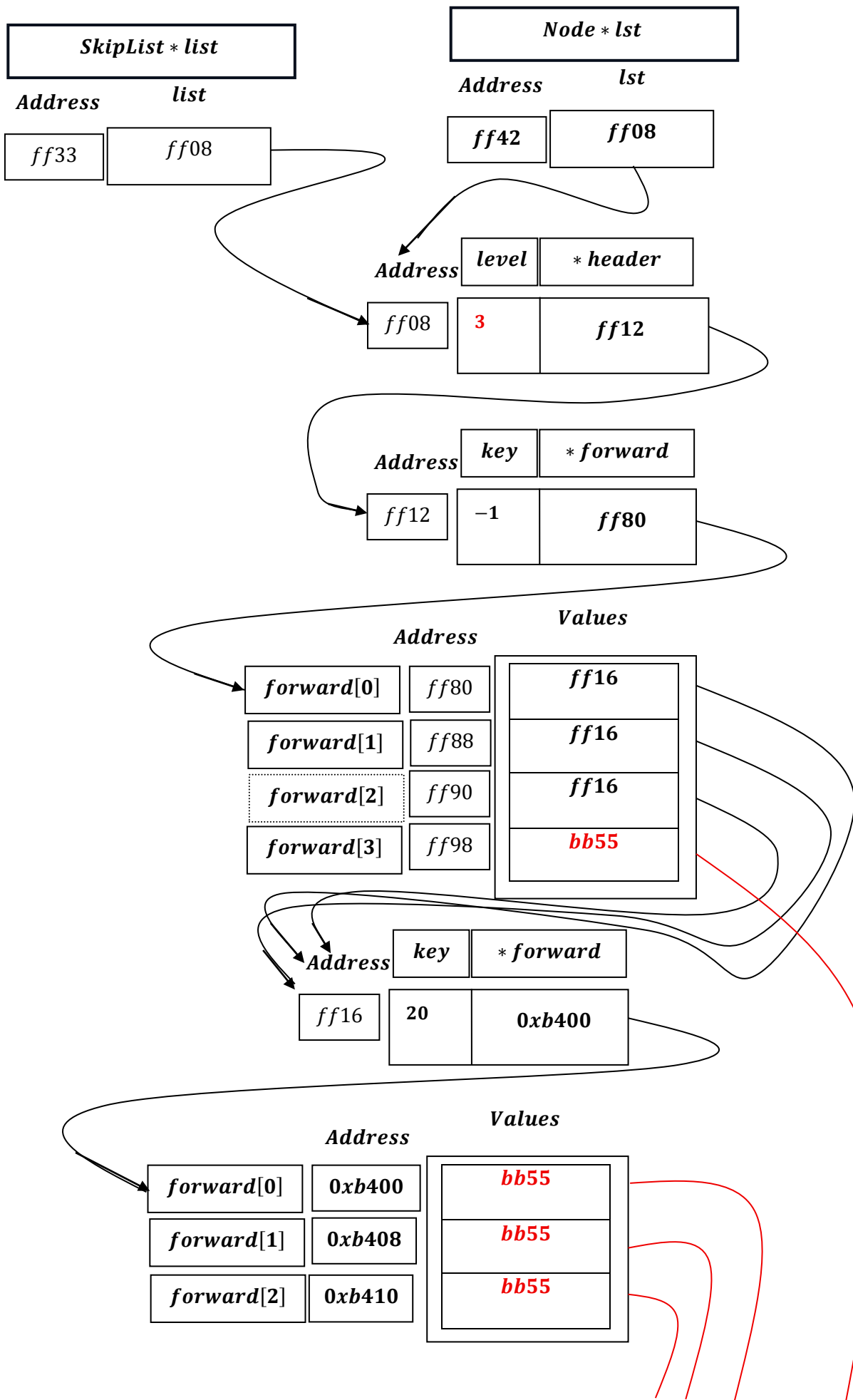
**$i = -1; i \geq 0 \left[ \begin{array}{l} \text{For loop condition false,} \\ \text{and Loop exits.} \end{array} \right]$**

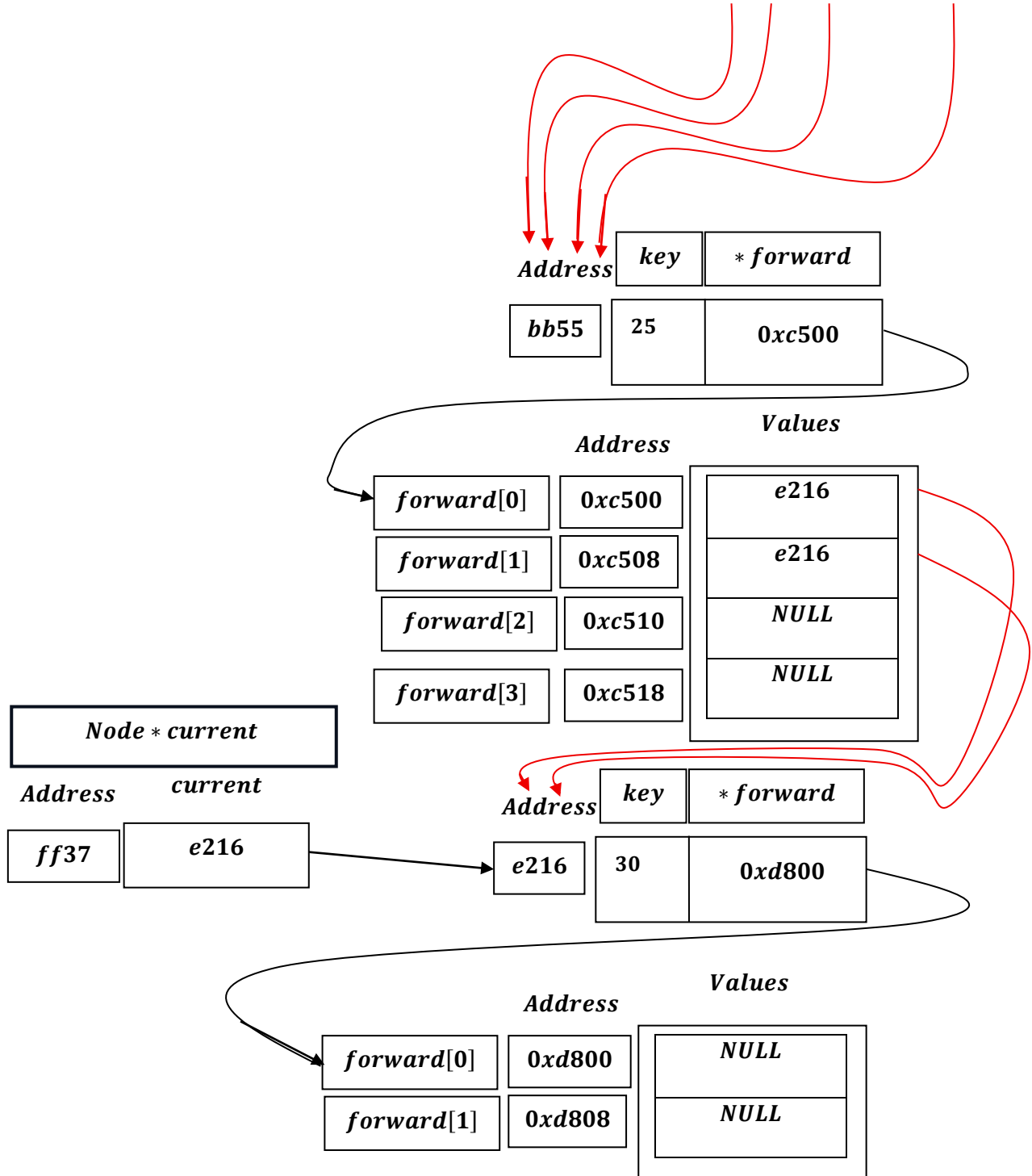




```
// Step into the exact match at level 0
current = current->forward[0];
```

*current* = *current*: bb55 → forward[0]: 0xc500 → e216.





```
if (current != NULL && current->key == key) {
    cout << "Found key " << key << " in the Skip List." << endl;
}
```

*current: e216*  $\neq$  *NULL* [True] AND *current: e216*  $\rightarrow$  *key: 30* = *key: 30* [True]:  
 Print "Found key 30 in the Skip List."

3<sup>rd</sup>, when  $current \rightarrow key \neq key$   $\begin{bmatrix} \text{Element} \\ \text{Not} \\ \text{Found} \end{bmatrix}$

```
searchElement(list, key);
```

*searchElement(list: ff08, key: 15) → passed .*

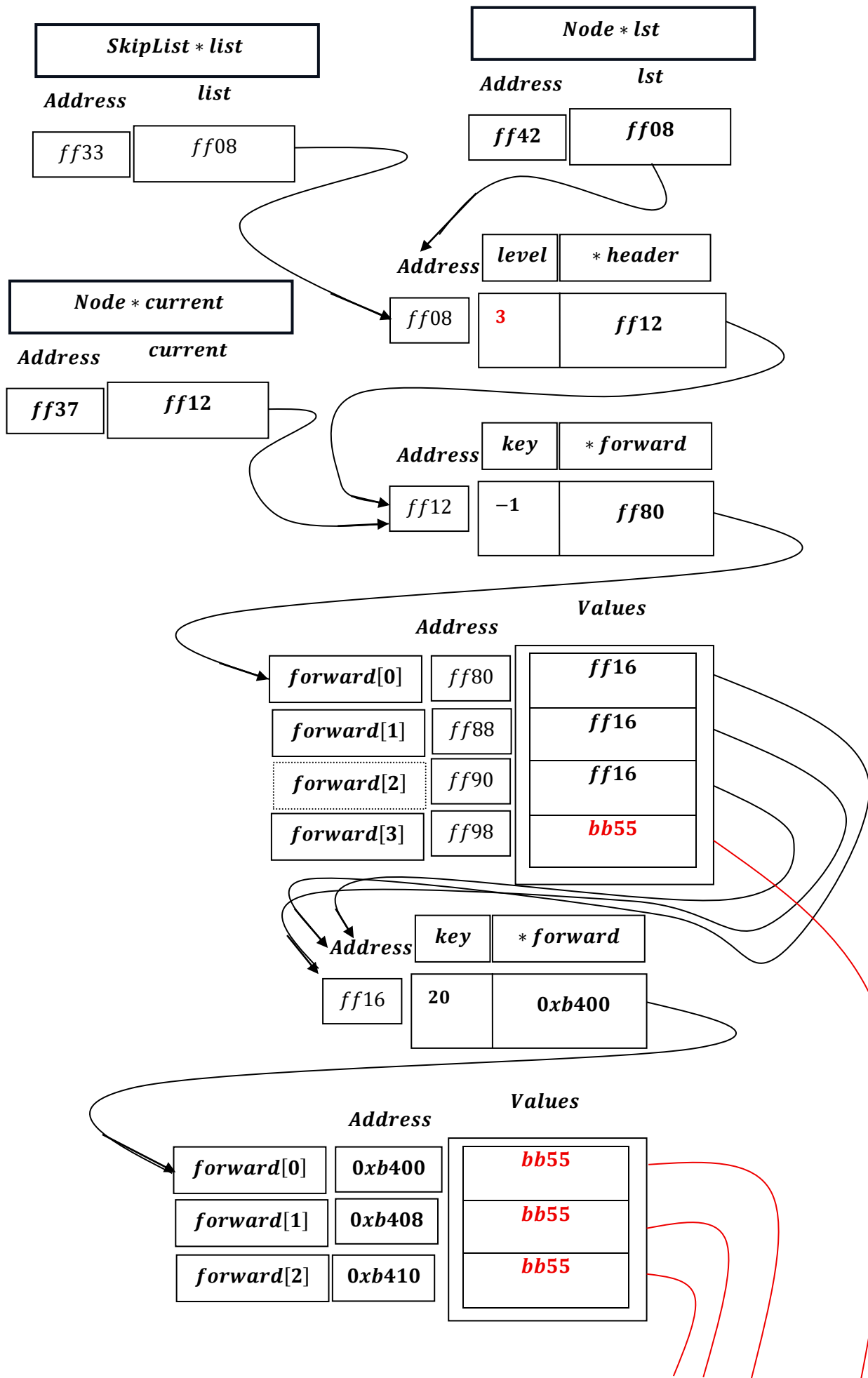
*And,*

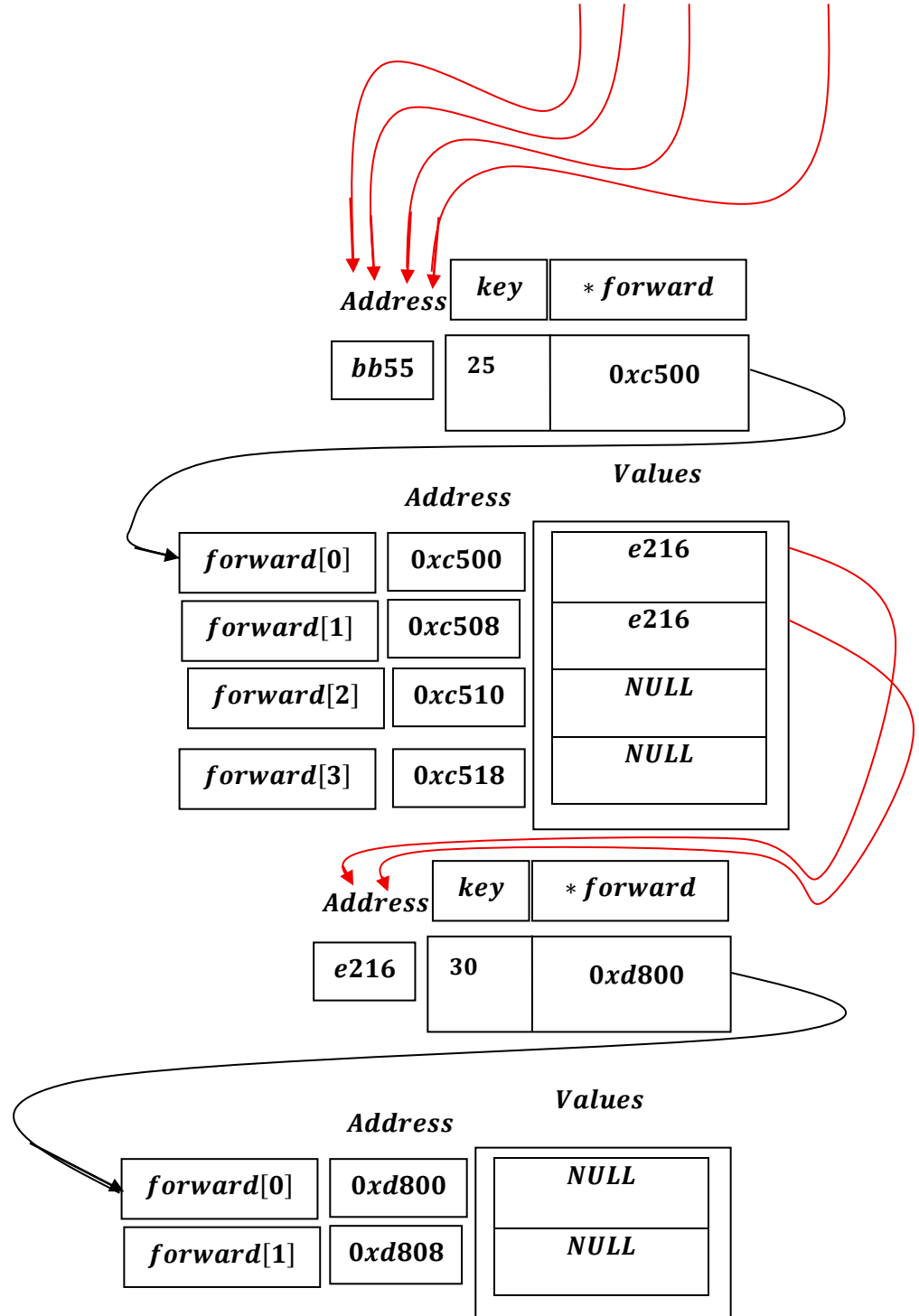
```
void searchElement(struct SkipList *lst, int key)
```

*searchElement(list := lst: ff08, key := key: 15) → Assigned*

```
struct Node *current = lst->header;
```

*current = lst → header = ff12.*





```

for (int i = lst->level; i >= 0; i--) {
    while (current->forward[i] != NULL && current->forward[i]->key <
key) {
        current = current->forward[i];
    }
}

```

*lst* → *level* = 3

$i = lst \rightarrow level = 3; i \geq 0$

**while:**  $current: ff12 \rightarrow forward[3]: ff98 \rightarrow bb55 \neq NULL$  [True] **AND**

$current: ff12 \rightarrow forward[3]: ff98[bb55] \rightarrow key: 25 < key: 15$ 

|                  |
|------------------|
| <b>Condition</b> |
| <b>False</b>     |
| <b>Loop</b>      |
| <b>Exit</b>      |

$i -- \Rightarrow i = i - 1 = 3 - 1 = 2$

$i = 2; i \geq 0$

**while:**  $current: ff12 \rightarrow forward[2]: ff90 \rightarrow ff16 \neq NULL$  [True] **AND**

$current: ff12 \rightarrow forward[3]: ff90[ff16] \rightarrow key: 20 < key: 15$ 

|                  |
|------------------|
| <b>Condition</b> |
| <b>False</b>     |
| <b>Loop</b>      |
| <b>Exit</b>      |

$i -- \Rightarrow i = i - 1 = 2 - 1 = 1$

**while:**  $current: ff12 \rightarrow forward[1]: ff88 \rightarrow ff16 \neq NULL$  [True] **AND**

$current: ff12 \rightarrow forward[1]: ff88[ff16] \rightarrow key: 20 < key: 15$ 

|                  |
|------------------|
| <b>Condition</b> |
| <b>False</b>     |
| <b>Loop</b>      |
| <b>Exit</b>      |

$i -- \Rightarrow i = i - 1 = 1 - 1 = 0$

**while:**  $current: ff12 \rightarrow forward[0]: ff80 \rightarrow ff16 \neq NULL$  [True] **AND**

$current: ff12 \rightarrow forward[0]: ff80[ff16] \rightarrow key: 20 < key: 15$ 

|                  |
|------------------|
| <b>Condition</b> |
| <b>False</b>     |
| <b>Loop</b>      |
| <b>Exit</b>      |

$i -- \Rightarrow i = i - 1 = 0 - 1 = -1$

$i = -1; i \geq 0$ 

|                                                      |
|------------------------------------------------------|
| <b>For loop condition false,<br/>and Loop exits.</b> |
|------------------------------------------------------|

```
// Step into the exact match at level 0
current = current->forward[0];
```

*current = current: ff12 → forward[0]: ff80 → ff16.*

```
if (current != NULL && current->key == key) {
    cout << "Found key " << key << " in the Skip List." << endl;
} else {
    cout << "Key " << key << " not found." << endl;
}
```

*current: ff16 ≠ NULL [TRUE] AND current: ff16 → key: 20 ≠ key: 15*  $\left[ \begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

**Else**

*Print "Key 15 not found."*

*And , function ends.*

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